

1. Where is the optimal placement for the database initialization code `ensure_users_table()` in a Flask application? Should it be at the top level of `app.py` or triggered upon the first request?

Gemini : Based on the best practices for deploying a **Flask** application to **AWS EC2**, here is the breakdown of where to place your database initialization code.

The "Gold Standard": Application Factory Pattern

The most professional approach is to use a dedicated database module and trigger the initialization within an **Application Context**.

- **Placement:** Inside a `with app.app_context():` block in your `app.py` (after creating the app instance but before calling `app.run()`).
- **Why it wins:** It ensures that variables like your **MySQL** credentials and Flask configurations are fully loaded before the table check runs.
- **Deployment Advantage:** When you move your code from your **Laptop** to **AWS EC2**, using an application factory makes it much easier to manage different environments (development vs. production) via **GitHub**.

Why the "Alternatives" Are Risky

1. **Top-level of `app.py`:**
 - **The Risk:** Code at the top level executes every time the module is imported. If you use a production server like Gunicorn on **EC2**, it might spin up multiple workers, each trying to run `ensure_users_table()` simultaneously, which is messy and inefficient.
2. **Triggered on first request:**
 - **The Risk:** Flask has officially deprecated the `before_first_request` decorator in newer versions. More importantly, your **Python Scraper** needs that database structure to exist *before* it starts saving bike occupancy data every 5 minutes.

Suggested Project Architecture

Since your project involves both a **Web App** and a **Web Scraper**, you should treat the database as a shared resource:

- **Database Module:** Create a `db_utils.py` that contains `ensure_users_table()` and other **MySQL** logic.
- **The Scraper:** Have your scraper call this function when it starts up so it doesn't crash if the tables are missing.
- **The Flask App:** Call it within the app context to ensure the **Web Application** is ready for user registration.

Pro Tip: Since you are using **GitHub** for development, make sure your database credentials (host, user, password) are stored in an `.env` file and **never** committed to your repository.

Since you're implementing the **Journey Planner** or **Login** as a feature, are you planning to use standard SQL queries for your user management, or are you looking into an ORM like Flask-SQLAlchemy?

2. How do I correctly structure `app.py` in an AWS EC2 environment to support multi-process execution with Gunicorn, rather than relying on the built-in Flask development server?

Gemini: To transition your Flask application from a local environment to a production-ready **AWS EC2** instance, you must shift from the single-threaded development server to a multi-process WSGI server like **Gunicorn**.

In a multi-process environment, each "worker" is a separate process with its own memory space. This requires specific structural changes to `app.py` to ensure resources like your **MySQL database** and **Machine Learning model** are handled efficiently.

1. Implement the Application Factory Pattern

Instead of a global `app` object, use a function to create and configure the app. This prevents side effects when Gunicorn imports your module.

Python

```
from flask import Flask
```

```
from .db_utils import ensure_users_table
```

```
from .ml_utils import load_model
```

```
def create_app():
```

```
    app = Flask(__name__)
```

```
    # Configuration and blueprints go here
```

```
    with app.app_context():
```

```
        # Initialize database tables once per deployment
```

```

ensure_users_table()

# Pre-load your ScikitLearn model into memory [cite: 5]

# Note: In multi-process, each worker loads its own copy

app.model = load_model()

return app

# Only for local testing

if __name__ == "__main__":

    app = create_app()

    app.run(host='0.0.0.0', port=5000)

```

2. Strategic Resource Management

When running multiple Gunicorn workers on **EC2**, keep these constraints in mind:

- **Database Connections:** Each worker process will open its own connections to **MySQL**. Ensure your MySQL server is configured to handle the total number of concurrent connections (Workers × Connections per pool).
- **Machine Learning Model Memory:** Your **ScikitLearn** model will be loaded into the RAM of every worker process. If your model is large, too many workers may exhaust your EC2 instance's memory.
- **Shared State:** Do not use global Python variables to store shared data (like a list of bikes). Use the **MySQL database** for any state that must be consistent across all processes.

3. Execution on AWS EC2

Once your code is deployed via **GitHub**, you start the server from the command line. Unlike the development server, you do not run `python app.py`. Instead, you point Gunicorn at your factory function:

```
gunicorn -w 4 -b 0.0.0.0:8080 "app:create_app()"
```

- **-w 4:** Spawns 4 worker processes. A common rule is $(2 \times \text{number of CPU cores}) + 1$.
- **-b 0.0.0.0:8080:** Binds the server to all network interfaces on port 8080.

4. Integration with the Web Scraper

Your **Python Scraper** runs as a completely separate process from the Flask web app, typically triggered by a **cron** job on the **EC2** instance every 5 minutes. Because they are separate processes, the Scraper and Gunicorn workers *only* communicate through the **MySQL database**.

Are you planning to use a reverse proxy like Nginx in front of Gunicorn, or will Gunicorn handle the incoming traffic directly?

3. Why didn't Git prompt for my credentials during the push?

Gemini: Since your project requires using **GitHub** for both development and deployment to **AWS EC2**, it is common for Git to handle authentication silently. If you weren't prompted for a username or password during your **git push**, it is likely due to one of the following reasons:

1. Git Credential Manager (GCM)

Most modern operating systems use a credential helper that automatically stores and retrieves your GitHub credentials after the first successful login.

- **Windows/macOS:** They typically have a built-in "Git Credential Manager" that stores your Personal Access Token (PAT) in the system keychain.
- **Result:** As long as the token is valid, you won't see a prompt until the token expires or is revoked.

2. SSH Key Authentication

If you configured the remote URL of your repository using the SSH format (e.g., **git@github.com:username/repo.git**) rather than HTTPS, Git uses **SSH keys**.

- **Process:** Your computer uses a private key to "shake hands" with GitHub's public key.
- **Result:** This process is encrypted and happens in the background without needing a password, provided your SSH agent is running.

3. Credential Caching

You may have previously configured Git to "remember" your credentials for a specific duration using a command like:

```
git config --global credential.helper cache
```

- This stores your credentials in memory for a short period (default is 15 minutes), allowing you to push multiple times without re-authenticating.

Critical Security Tip for your Project

As you move into **Sprint 3 (Frontend)** and **Sprint 4 (ML Model)**, keep the following in mind:

- **Avoid Hardcoding:** Never push files to **GitHub** that contain your **MySQL** passwords or **OpenWeather** API keys.
- **Use .gitignore:** Ensure your environment variables and local database configurations are ignored by Git. This is essential for maintaining a "stable communication channel" and secure deployment on **AWS**.

Are you currently using the **HTTPS** or **SSH** URL for your remote repository? (You can check by running `git remote -v`).

4. After updating code via `git pull` on EC2, what is the most efficient command to force a synchronization and gracefully restart Gunicorn processes without manually killing PIDs?

Gemini: To ensure your **Flask** application on **AWS EC2** reflects the latest changes from **GitHub** without dropping active user connections, you should use signals or a process manager rather than manual PID hunting.

1. The Pro Way: Systemd (Recommended)

If you followed standard production setup for your bike-sharing app, you likely configured Gunicorn as a **systemd** service. This is the most efficient way to manage restarts.

- **Command:** `sudo systemctl restart gunicorn`
- **Why:** Systemd handles the shutdown and restart logic automatically, ensuring the environment variables and paths defined in your service file are reapplied.

2. The "Graceful Reload" Signal

If you are running Gunicorn directly or via a script, use the **HUP (Hangup)** signal. This tells the Gunicorn master process to spawn new workers using the updated code and then gracefully kill off the old workers only after they finish processing current requests.

- **Command:** `kill -HUP $(pgrep gunicorn | head -1)`
- **How it works:**
 - `pgrep gunicorn` finds the process IDs.
 - `head -1` targets the master process.
 - `-HUP` triggers the reload without stopping the master process itself.

3. Verification & Troubleshooting

After performing the restart, verify that the new worker processes are running and that your **ScikitLearn** model or updated **REST Interface** is functioning correctly:

- **Check Status:** `sudo systemctl status gunicorn`
- **View Logs:** `journalctl -u gunicorn -f`

- **Sync check:** Since your project requires a **MySQL Database** connection , ensure that your new code hasn't broken the connection string or initialization logic you set up in **Sprint 2**.

Summary Table for EC2 Management

Goal	Command	Impact
Quick Sync	<code>git pull origin main</code>	Updates local files on EC2.
Graceful Restart	<code>kill -HUP <PID></code>	Reloads code; zero downtime.
Full Restart	<code>sudo systemctl restart gunicorn</code>	Restarts everything; best for config changes.
Monitor	<code>tail -f /var/log/gunicorn/error.log</code>	Real-time debugging of the Flask app.

By using these commands, you maintain the "stable communication" and professional development workflow required for the **COMP30830 project**.

5. While migrating MySQL from local to EC2, I encountered a GTID-related export error. How can I successfully import the SQL file without requiring elevated database permissions?

Gemini: This GTID error is a classic "rite of passage" when migrating from a local environment to a cloud environment like **AWS EC2**. The error happens because `mysqldump` includes Global Transaction Identifier (GTID) metadata by default, and your target database user on EC2 lacks the **SUPER** privileges required to set these global variables.

To successfully import your **MySQL** data without needing elevated permissions, use one of the following methods:

1. The Cleanest Fix: Re-export without GTIDs

The most efficient way is to generate a new export file that specifically tells MySQL to ignore GTID information. Run this command on your **local laptop**:

Bash

```
mysqldump --set-gtid-purged=OFF -u [username] -p [database_name] > export_file.sql
```

- **--set-gtid-purged=OFF**: This is the magic flag. It prevents the inclusion of **SET @@GLOBAL.GTID_PURGED** in the SQL file, allowing a standard user to perform the import.

2. The Quick Fix: Edit your existing SQL file

If you cannot re-export the data, you can manually strip the problematic lines from your existing **.sql** file using a text editor or a command-line tool like **sed**:

Bash

```
# This removes the GTID_PURGED lines and the associated comments
```

```
sed -i '/@@GLOBAL.GTID_PURGED/d' export_file.sql
```

Integration with your COMP30830 Project

As you move through **Sprint 3** and transition from **Local MySQL** to **EC2**:

- **Database Setup**: Ensure your target database on EC2 has the same schema name as your local one to avoid pathing issues in your **Flask App**.
- **Security Groups**: When you perform the import on EC2, ensure your **AWS Security Group** allows traffic on port **3306** if you are connecting remotely, or perform the import locally on the EC2 instance.
- **Consistency**: Since your **Web Scraper** will be writing to this database every 5 minutes, verifying that the tables (like those created by **ensure_users_table()**) are correctly imported is vital for the **ML Model** to access historical data later.

Summary of Commands

Step	Action	Command
Export	Generate clean SQL	<code>mysqldump --set-gtid-purged=OFF ...</code>
Transfer	Move to EC2	<code>scp export_file.sql ec2-user@your-ec2-ip:~</code>

Import	Execute on EC2	<code>mysql -u [user] -p [db_name] < export_file.sql</code>
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By removing the GTID requirement, you bypass the need for **SUPER** privileges, allowing you to finalize the migration and move on to the **Frontend** development.

6. My Flask project runs perfectly locally but returns 404 errors for all CSS and image files after deployment to EC2. What are the typical causes for this, and how can I implement a robust `base_dir` dynamic pathing strategy in `app.py`?

Gemini: This is a classic deployment "gotcha" when moving from a development environment to a production Linux environment like **AWS EC2**. Locally, your OS might be forgiving, but EC2 is strictly case-sensitive and relies on specific path resolutions for the **HTML/JS + Flask** stack.

Typical Causes for 404 Errors on EC2

- **Case Sensitivity:** Linux is case-sensitive. If your code references `style.css` but the file is named `Style.css`, it will work on Windows/macOS but return a 404 on EC2.
- **Hardcoded Absolute Paths:** If your `app.py` or HTML templates use local paths like `C:\Users\Project\static`, these will obviously fail in the Linux file system of an **EC2** instance.
- **Nginx Configuration:** If you are using Nginx as a reverse proxy, it often tries to serve static files itself for efficiency. If the `root` or `alias` path in your Nginx config is incorrect, it won't find your CSS or images.
- **Working Directory issues:** When running via **Gunicorn**, the "current working directory" might not be what you think it is, causing Flask to look in the wrong place for the `/static` folder.

Implementing a Robust Dynamic Pathing Strategy

To make your **bike sharing** web application portable between your laptop and EC2, you should never hardcode paths. Instead, define a `BASE_DIR` at the top of your `app.py`.

Step 1: Update `app.py`

Use the `os` module to detect the absolute path of the directory where `app.py` resides.

Python

```
import os
```

```
from flask import Flask
```

```
# Dynamically find the absolute path to the project root
```


This works regardless of whether you are on Windows or AWS EC2

```
base_dir = os.path.abspath(os.path.dirname(__file__))
```

```
app = Flask(__name__,
```

```
    static_folder=os.path.join(base_dir, 'static'),
```

```
    template_folder=os.path.join(base_dir, 'templates'))
```

Step 2: Use `url_for` in Templates

Never reference your CSS or images with `/static/style.css`. Use Flask's built-in helper to ensure the paths are generated correctly by the **REST Interface**.

HTML

```
<link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}">
```

```

```

Pro-Tips for your Bike Sharing Project

- **Permissions:** Ensure the user running **Gunicorn** has read permissions for your static folder. You can test this on EC2 with `ls -l static/`.
- **Static Mapping:** If you are using Nginx, ensure your `/etc/nginx/sites-available/default` file includes a location block for static files that points to your dynamic `base_dir` location:
- Nginx

```
location /static/ {
```

```
    alias /home/ubuntu/your_project/static/;
```

```
}
```

- **Testing:** Part of your development process includes **Testing**. Always do a hard refresh (Ctrl+F5) in your browser on EC2 to ensure you aren't seeing cached 404s.

Since you're working on the **Frontend** and **Flask App** components right now, would you like help configuring your Nginx server block to specifically handle these static files more efficiently?

7. Our backend scraping script on EC2 occasionally terminates due to SSH disconnections or system timeouts. How can I implement a Systemd service or a daemon process to ensure 24/7 continuous data collection?

Gemini: For a project that requires collecting data every 5 minutes and every hour, relying on an active SSH session is the most common point of failure. On **AWS EC2**, the professional way to ensure your **Python Scraper** runs continuously is to configure it as a **Systemd Service**.

Why Systemd?

- **Automatic Restart:** It will restart your scraper automatically if it crashes due to a network error or memory leak.
 - **Persistence:** The process starts automatically when the EC2 instance boots up.
 - **Decoupling:** It runs in the background, so you can safely disconnect your SSH session.
 - **Logging:** All output (stdout/stderr) is captured and can be viewed using [journalctl](#).
-

Step-by-Step Implementation

1. Create the Service File

Open a new service file on your EC2 instance:

```
sudo nano /etc/systemd/system/bike_scraper.service
```

2. Define the Configuration

Paste the following configuration, adjusting paths to match your project structure:

Ini, TOML

[Unit]

Description=Dublin Bikes Web Scraper Service

After=network.target mysql.service

[Service]

User running the script (usually ubuntu or ec2-user)

User=ubuntu

WorkingDirectory=/home/ubuntu/your_project_folder

Path to your virtual environment's python executable

```
ExecStart=/home/ubuntu/your_project_folder/venv/bin/python scraper.py
```

```
# Ensures the service restarts after a crash
```

```
Restart=always
```

```
RestartSec=5
```

```
# Logs output to systemd
```

```
StandardOutput=journal
```

```
StandardError=journal
```

```
[Install]
```

```
WantedBy=multi-user.target
```

3. Enable and Start the Service

Run these commands to register and launch your background data collector:

- **Reload Systemd:** `sudo systemctl daemon-reload`
- **Enable on Boot:** `sudo systemctl enable bike_scraper.service`
- **Start Now:** `sudo systemctl start bike_scraper.service`
- **Verify Status:** `sudo systemctl status bike_scraper.service`

Troubleshooting & Logs

Since your scraper is storing data in a **MySQL Database**, you need to monitor the logs to ensure the connections aren't timing out. Use the following command to see real-time output:

```
journalctl -u bike_scraper.service -f
```

Architectural Integration

In the **COMP30830 architecture**, the **Web Scraper** and **Flask App** are separate processes. While the Flask app is managed by Gunicorn, the Scraper is a standalone daemon feeding the **historical data** into your database.

Does your current scraping script use a `while True` loop with a `time.sleep()` delay, or are you planning to trigger it via a Systemd Timer for more precise scheduling?

8. For the bike prediction model, should we use the large-scale historical dataset provided by the instructor or the real-time data we scraped ourselves over the past two weeks?

Gemini: For the **ML Model Development** phase of your COMP30830 project, you should use the **large-scale historical dataset (quarterly data) provided by the instructor** to train your prediction model.

However, your project architecture requires you to use **both** datasets for different specific purposes:

1. The Instructor-Provided Historical CSV

- **Purpose:** This dataset is specifically designated for **ML Model Development**.
- **Training:** You will use this quarterly data in a Jupyter Notebook with ScikitLearn to create your `.pkl` model file.
- **Reasoning:** Two weeks of scraped data is insufficient to capture complex patterns such as how demand varies by the hour of the day or the day of the week. The provided historical data covers a much longer period, allowing the model to learn more robust trends.

2. Your Real-Time Scraped Data

- **Purpose:** This data is used for the **live features** of your web application.
- **Current Occupancy:** You must display the most recent data on the Google Map to show users how many bikes are currently available at each station.
- **Station Charts:** When a user clicks a station, you use your locally stored data (aiming for at least two days of history) to display detailed hourly and daily occupancy bar charts.

Summary of Dataset Roles

Feature	Dataset to Use	Source
Model Training (ScikitLearn)	Instructor's <code>.csv</code> (Historical Quarterly Data)	
Current Map Markers	Real-time JCDecaux Scraped Data	
Detailed Occupancy Charts	Locally Stored Scraped Data (1-2 days)	
Weather Forecast Display	OpenWeather Scraped Data (Every hour)	

According to your schedule, you should have already set up your **Web Scraper** in Sprint 1 , while the **ML Model** training happens later in Sprint 4.

Are you planning to use a **Regression** model to predict the exact number of bikes, or a **Classification** model to predict categories like "High" or "Low" availability?

9. If we merge them, will the distribution mismatch cause significant bias when predicting the next 24 hours?

Gemini: Yes, merging them blindly will likely cause **Concept Drift**, leading to significant bias in your 24-hour predictions. Here is why this mismatch happens and how to handle it for your project:

The Source of the Bias

- **Temporal Gap:** The instructor-provided quarterly data reflects historical patterns (with some charts in your brief dating back to 2017).
- **Infrastructure Changes:** Since the system launched in 2009, the number of stations and bikes has changed. If a station was added or expanded in 2026, the old data won't reflect its current capacity.
- **Behavioral Shifts:** While demand still varies by hour and day of the week, the *intensity* of those peaks may have shifted due to new cycling lanes or changes in city commuting habits between the historical period and 2026.

Why "Merging" is Risky

If you simply combine the 2017-era quarterly data with your 2026 scraped data into one training pool:

- **Diluted Recency:** The high volume of historical data will "drown out" the recent patterns you've scraped over the last two weeks.
- **Static Bias:** The model might learn that a station "usually" has 10 bikes based on old data, even if your 2026 scraping shows it now rarely exceeds 5 bikes due to higher demand.

Recommended Strategy for COMP30830

To minimize bias while predicting the next 24 hours:

1. **Feature Engineering over Merging:** Use the historical dataset to train the model on **relative patterns** (e.g., "occupancy usually drops by 30% on Monday mornings") rather than absolute numbers.
2. **Input Current State:** When making a 24-hour prediction, your ScikitLearn model should take the **most recent scraped data** as a starting feature (input). This anchors the prediction in 2026 reality while using historical trends to "shape" the next 24 hours.
3. **Validation:** Use your two weeks of scraped data as a **test set** to see how well the model trained on historical data actually performs in the current 2026 environment.

Are you planning to include **weather data** (temperature/rain) from OpenWeather as a feature in your model, as it often explains sudden deviations from historical trends?

10. In our dataset, is the **minute** field actually beneficial for predicting bike availability, or should I transform it into coarser time intervals (e.g., 15 or 30-minute bins) to reduce the model's sensitivity to noise?

Gemini: For your **bike-sharing prediction model**, transforming the **minute** field into coarser intervals (15 or 30-minute bins) is generally the more effective strategy.

While your **Web Scraper** collects data every 5 minutes, using raw minutes in a **ScikitLearn** model often introduces "noise" without providing meaningful predictive power.

Why Coarser Bins (15/30 mins) Are Beneficial

- **Noise Reduction:** Bike returns and rentals are stochastic; a single person returning a bike at 8:04 versus 8:09 doesn't represent a change in trend, but raw minute data might cause the model to overfit to these random fluctuations.
 - **Alignment with Human Behavior:** Commuting patterns and demand vary significantly by the hour of the day and day of the week. People generally plan trips in blocks (e.g., "leaving for work at 8:30"), making 15 or 30-minute intervals more representative of actual user intent.
 - **Reduced Model Complexity:** Using raw minutes (0–59) creates higher cardinality for the time feature. Binning simplifies the feature space, helping your **ML model** generalize better across the historical dataset.
-

Project-Specific Context

- **UI Consistency:** The project requirements specifically mention displaying occupancy "broken down by hour" for the station charts. Aligning your model's intervals with these display increments ensures a more cohesive user experience.
- **Data Density:** Your script aims to have two days of locally downloaded data at any time. With a 5-minute collection cycle, a 30-minute bin still provides 48 distinct data points per day, which is more than enough to capture the "Hourly Occupancy statistics" requested in the brief.

Implementation Tip

When you process the historical **.csv** provided by the instructor, you can create a new feature like **time_slot** using: `df['time_slot'] = df['hour'] + (df['minute'] // 30) * 0.5`

This creates a continuous numerical feature that captures both the hour and the half-hour block, which is very effective for regression-based models.

Are you planning to use **One-Hot Encoding** for these time bins, or will you treat the time of day as a continuous cyclical feature (e.g., using sine/cosine transforms)?

11. **XGBoost theoretically outperforms Random Forest on tabular data, but why is it performing worse on our specific dataset (MAE 5.4 vs. 0.98)? Is it due to high feature correlation, or has our data scale not yet reached the threshold where gradient boosting's advantages become apparent?**

Gemini: The discrepancy you're seeing (\$MAE = 5.4\$ vs. \$MAE = 0.98\$) is quite dramatic for a system where many stations only have a capacity of roughly **30–40 stands**. While XGBoost is often the "king" of tabular data, its poor performance here is likely due to a combination of **overfitting on a small scale** and **sensitivity to noise**.

Here is why your Random Forest (RF) is currently winning:

1. The "Scale Threshold" and Overfitting

Gradient Boosting Machines (GBM) like XGBoost are "greedy" learners that minimize a loss function sequentially. If you are training primarily on the **two weeks of scraped data** rather than the **large-scale historical quarterly dataset**, XGBoost is likely "memorizing" the noise in those specific two weeks.

- **Random Forest (Bagging):** Reduces variance by averaging multiple independent trees. It is much more robust to the "noisy" nature of bike-sharing data, where a single random group of tourists can skew occupancy at a specific station.
- **XGBoost (Boosting):** Reduces bias by focusing on previous errors. With a small dataset, it can become hyper-fixated on outliers, leading to a much higher Mean Absolute Error (MAE) on your test set:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

2. Sensitivity to Hyperparameters

Random Forest is famous for working exceptionally well "out of the box." In contrast, XGBoost is highly sensitive to its parameters. If you haven't tuned the **learning rate** (η), **max_depth**, or **min_child_weight**, XGBoost can easily diverge or overfit.

- A \$MAE\$ of **5.4** suggests the model is essentially guessing the "average" occupancy poorly, while **0.98** suggests the RF has successfully captured the core temporal patterns of the **hour** and **day of the week**.

3. Feature Correlation & Redundancy

You asked about high feature correlation. While tree-based models generally handle multicollinearity well, XGBoost's feature selection is more aggressive than Random Forest's.

- If your **weather data** (temperature/rain) and **temporal bins** are highly correlated with certain station IDs, XGBoost might be ignoring useful features in favor of a single "strong" but noisy predictor.
- Random Forest selects a **random subset of features** for every split, ensuring that "weaker" but still valid predictors (like small weather changes) still get a vote in the final prediction.

4. Data Distribution (Concept Drift)

The project brief highlights that demand varies significantly by **hour** and **day**. If the distribution of your two-week scraped data is even slightly different from the test set (e.g., one week was unseasonably rainy), XGBoost will struggle to generalize those "new" patterns compared to the more stable RF.

Recommendations for your Project

- **Prioritize the Historical CSV:** Ensure you are using the instructor-provided **quarterly data** for training. This provides the volume needed for XGBoost to actually "learn" rather than "memorize."
- **Check for Data Leakage:** An **MAE** of **0.98** is almost *too* good for bike-sharing data. Double-check that your "Available Bikes" target isn't accidentally included in your features (like "Available Stands").
- **Feature Engineering:** Instead of raw minutes, stick to the **15/30-minute bins** we discussed to further stabilize the input for the GBM.

Since you've achieved a **MAE** of **0.98** with Random Forest, you have a very strong baseline for your **Sprint 4 (ML Model)** report. Have you tried performing a Feature Importance plot to see which variables your RF is relying on most?

12. When predicting bike counts, why is it necessary to introduce RMSE as a secondary metric alongside MAE? What is its practical engineering significance in penalizing 'outlier errors' that might result in a poor user experience (e.g., a user arriving at an empty station)?

Gemini: In a high-stakes **bike-sharing interface**, **MAE (Mean Absolute Error)** and **RMSE (Root Mean Square Error)** serve different purposes. While MAE tells you the "typical" performance of your **ML model**, RMSE acts as a "risk assessment" tool that specifically targets catastrophic failures.

1. The Mathematical "Penalty" for Failure

The core difference lies in how each metric handles the magnitude of the error:

- **MAE (Linear):** Measures the average magnitude of errors without considering direction. If your model is off by 1 bike at 10 stations, the MAE is 1.0.
- **RMSE (Quadratic):** Squares the errors before averaging them. If your model is off by 1 bike at 9 stations but off by 10 bikes at the 10th station, the RMSE will be significantly higher than the MAE.
 - Formula:
$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

By squaring the error term $(y_i - \hat{y}_i)^2$, RMSE disproportionately penalizes large discrepancies. From an engineering perspective, a high RMSE relative to your MAE

indicates that your model is prone to "large misses," even if it is generally accurate on average.

2. Engineering Significance: The "Empty Station" Disaster

In the context of the **Journey Planner** or the **Dublin Bikes map**, not all errors are created equal.

- **Low Impact Error:** Your model predicts 5 bikes, but there are actually 4. The user can still rent a bike, and the **user experience** remains positive.
- **High Impact (Outlier) Error:** Your model predicts 5 bikes, but there are actually 0. The user follows the **Plan Your Journey** recommendation, arrives at the station, and finds it empty.

RMSE is necessary because it treats a single 10-bike error as much worse than ten 1-bike errors. In **Web application** engineering, this forces the model-building process to prioritize reliability. If your RMSE is high, your **prediction capabilities** are unreliable for real-world navigation because you are likely to mislead users during peak demand or at critical station junctions.

3. Practical Metrics for your Project Report

When you present your findings for **Sprint 4**, you should use the relationship between these two metrics to describe your model's stability:

Metric Relationship	Interpretation for Bike Sharing
RMSE $\approx$ MAE	The errors are consistent and small. The Journey Planner is highly reliable.
RMSE $\gg$ MAE	The model has a low "typical" error but occasionally makes massive mistakes (e.g., predicting 10 bikes when there are 0).

Using RMSE ensures that your **ScikitLearn** model isn't just "good on average" but is actively discouraged from making the specific outlier errors that lead to users arriving at empty stands after a long walk.

Since your **RMSE** will likely be higher during peak commute hours (as seen in the **Daily Occupancy Statistics**), have you considered evaluating these metrics separately for "Weekday Rush Hour" vs. "Weekends"?

